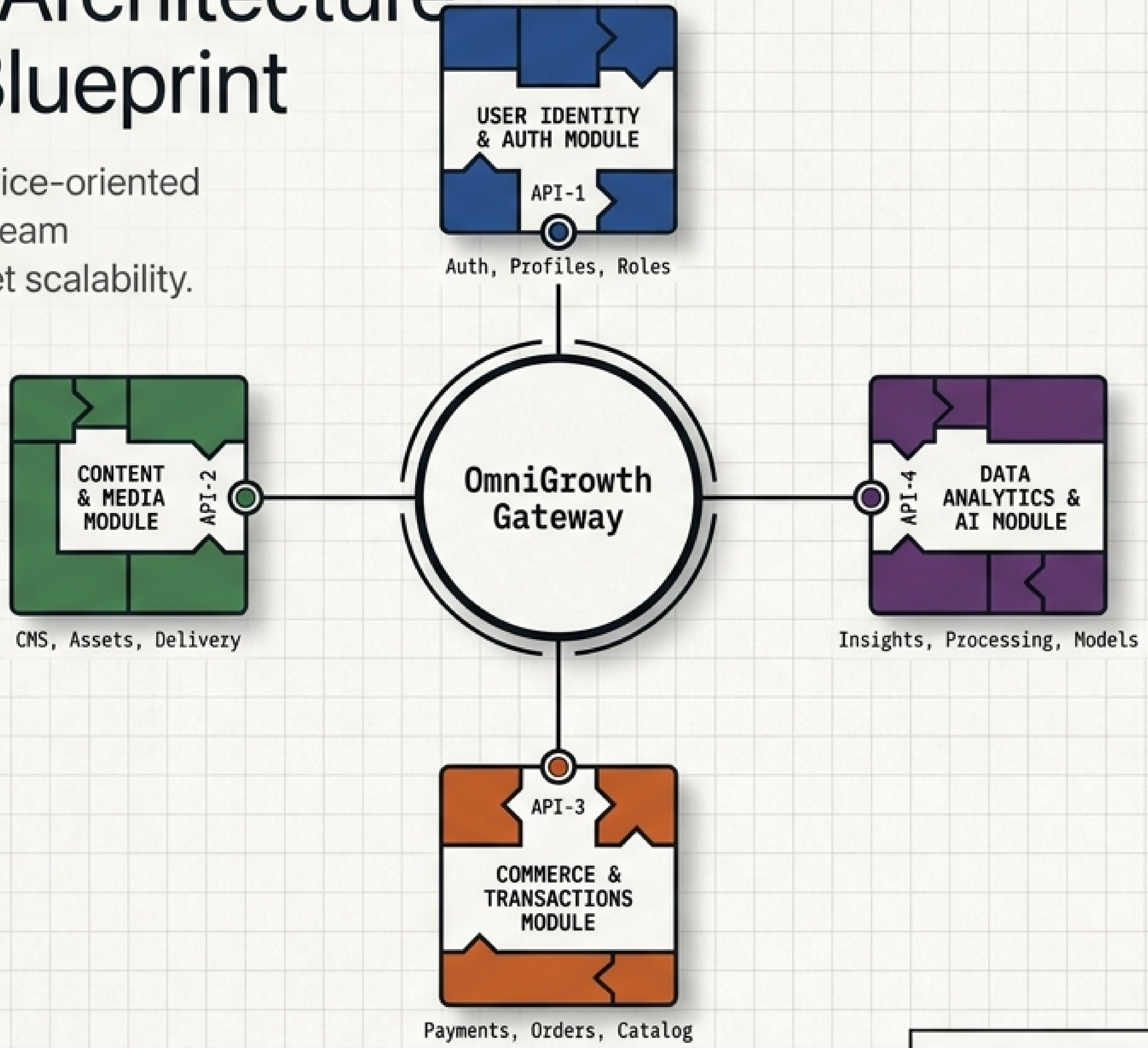
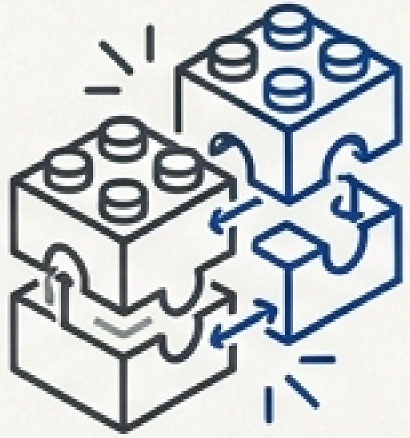
OmniGrowth Architecture & Technical Blueprint

A strictly decoupled, microservice-oriented system designed for absolute team independence and zero-budget scalability.



SYS-VER: 1.0 | STATUS: BASELINE DRAFT | ENV: ZERO-COST LOCAL

Architecture Driven by Two Non-Negotiable Mandates



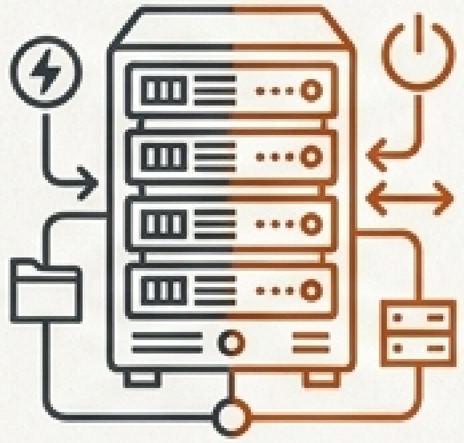
Team Independence

NFR-TEAM-001

The architecture cannot rely on a fixed developer headcount. Boundaries are drawn by business capability, not team size, allowing seamless scaling from 1 to N developers without overlapping codebases or undocumented knowledge dependencies.

API

API



Zero-Cost Environment

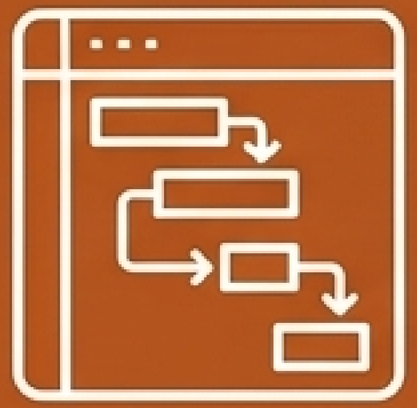
NFR-COST-001

Core development and demonstration require zero paid subscriptions. The baseline mandates free/open-source tooling, Docker for local execution, and a free/local path for all AI capabilities.

The Four Isolated Business Ecosystems

Administrative Hub

Controls identity, RBAC, and administrative audit context across the system.



Project & Process Orchestration

Drives task lifecycle, state transitions, and integration hooks for daily work.

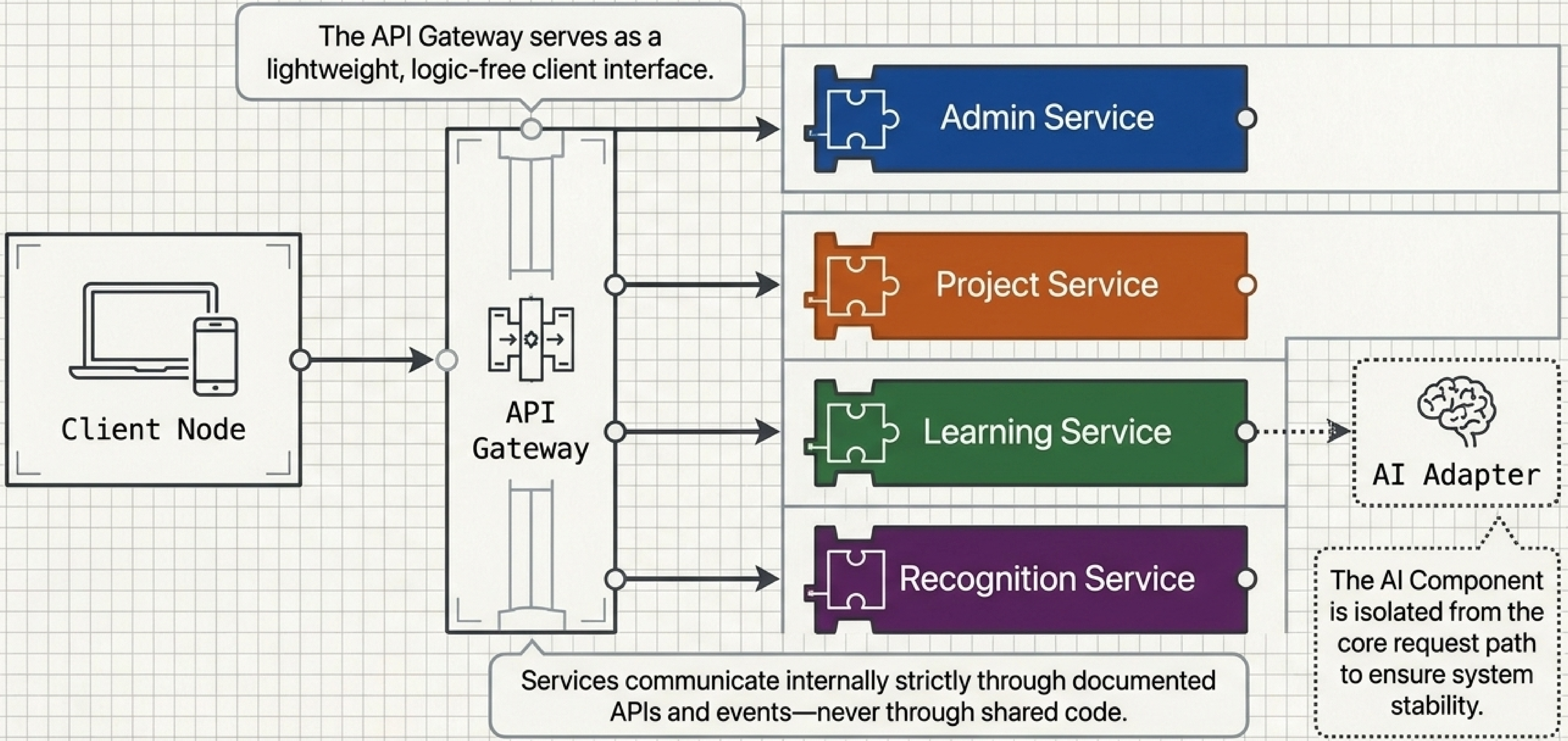
Holistic Learning Framework

Manages professional development across Physical, Social, Mental, and Spiritual dimensions.

Recognition & Incentives

Aggregates milestones into an idempotent, transactional point and reward ledger.

Enforcing Strict Isolation at the Microservice Layer



Bounded Contexts Replace Tangled Database Schemas

NFR-DAT-001

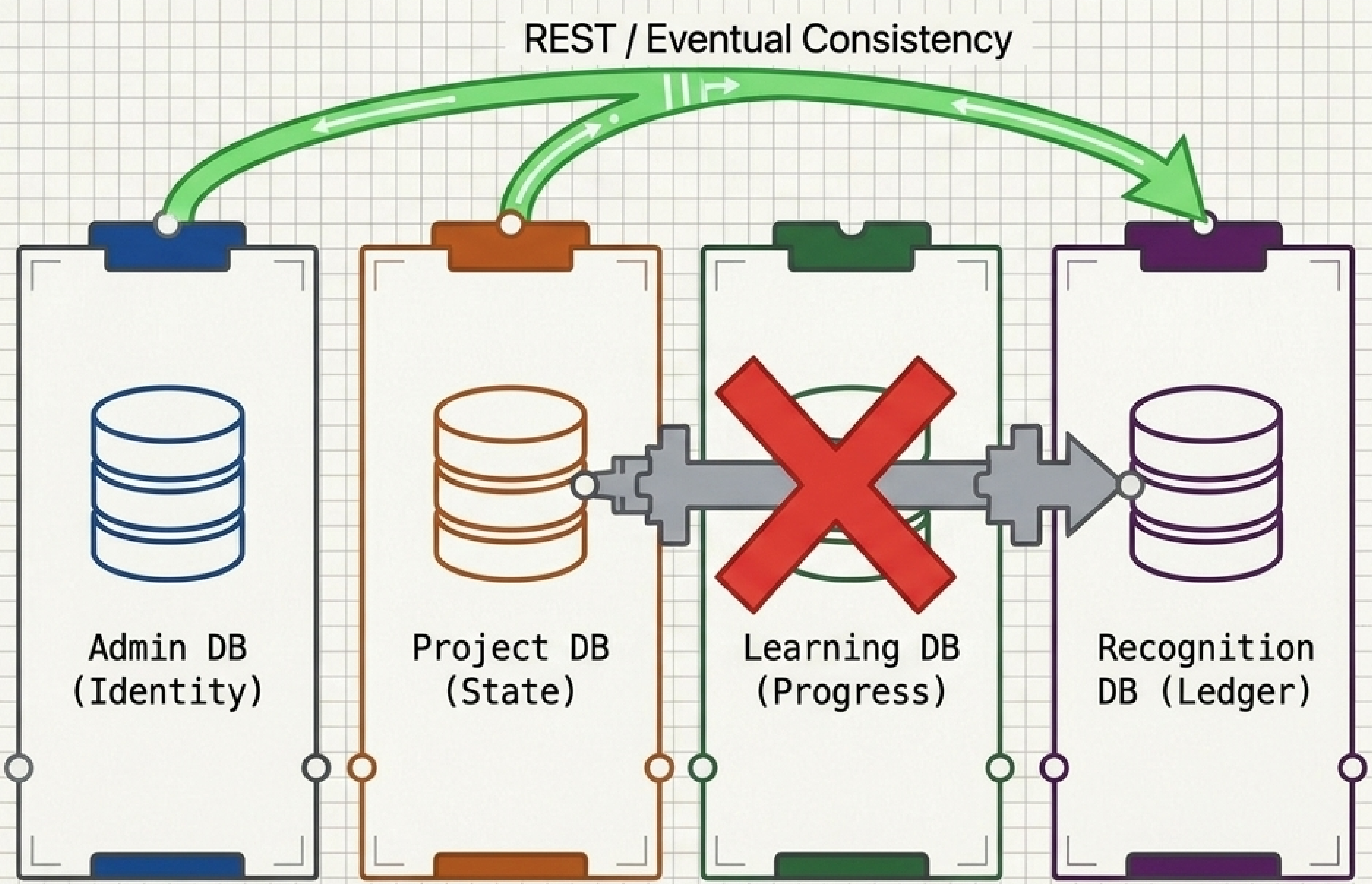
Clear data ownership per service. No direct cross-service DB reads or writes.

Identity references

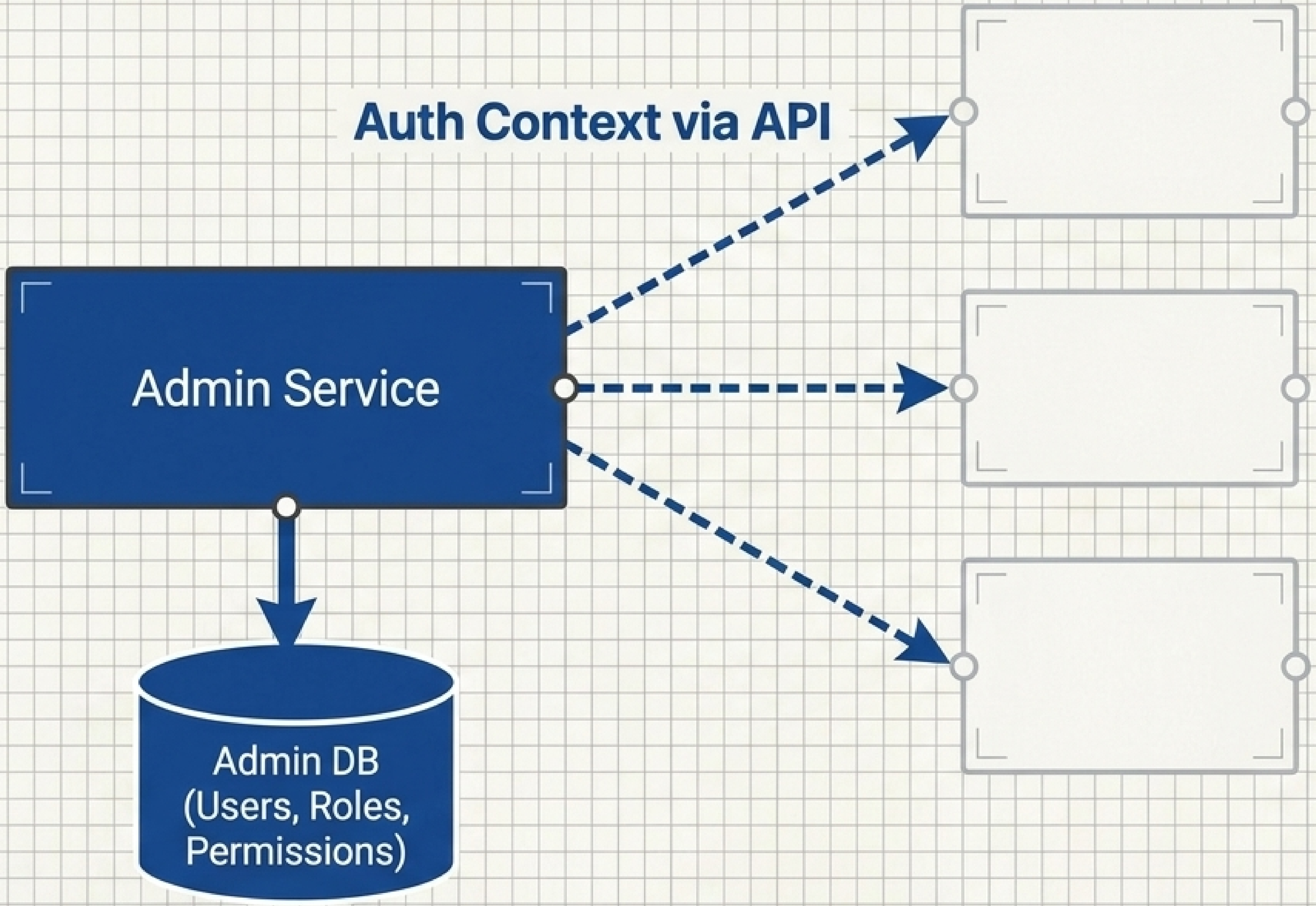
Downstream services store stable user IDs provided by the Admin Hub, not duplicated user tables.

Business Rules

Remain strictly within the owning service's database boundary.



Administrative Hub Acts as the Identity Foundation



Core Mandate

Manage user accounts and Role-Based Access Control (RBAC).

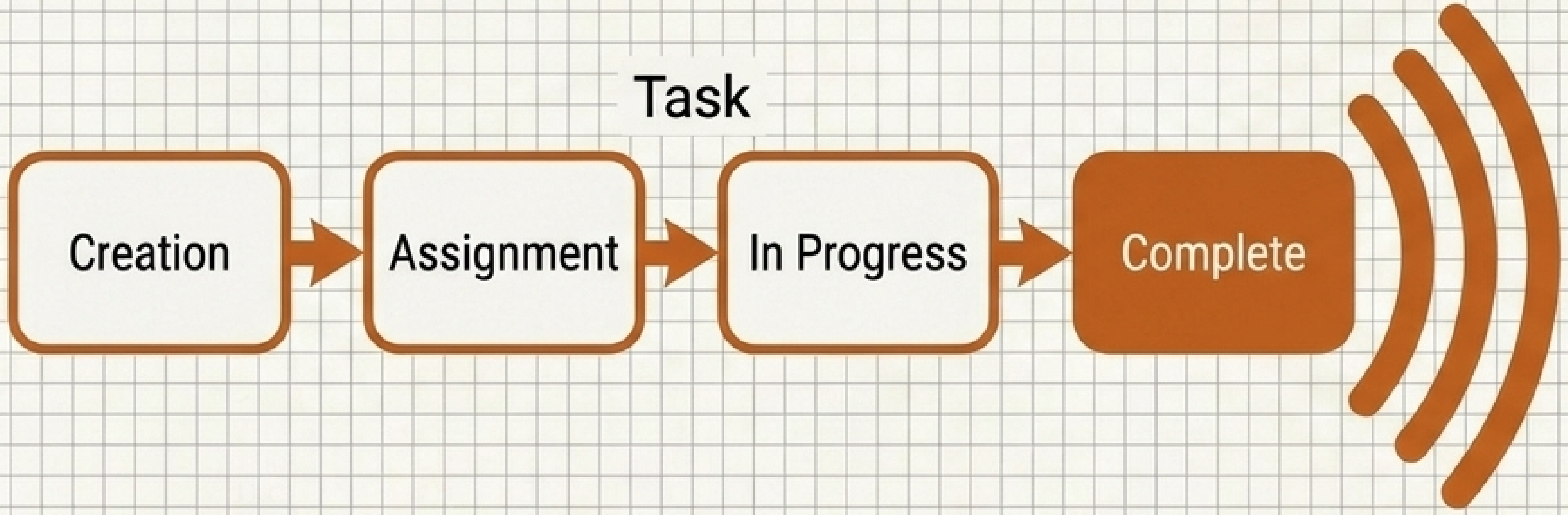
System Rule

Users cannot self-grant privileges. Other services do not modify access-control data.

Integration (FR-ADMIN-005)

Other services obtain identity exclusively through defined interfaces, ensuring centralized auditability for security-sensitive actions.

Project Orchestration Drives Activity States



Core Mandate

Manage projects, task dependencies, and workflow states.

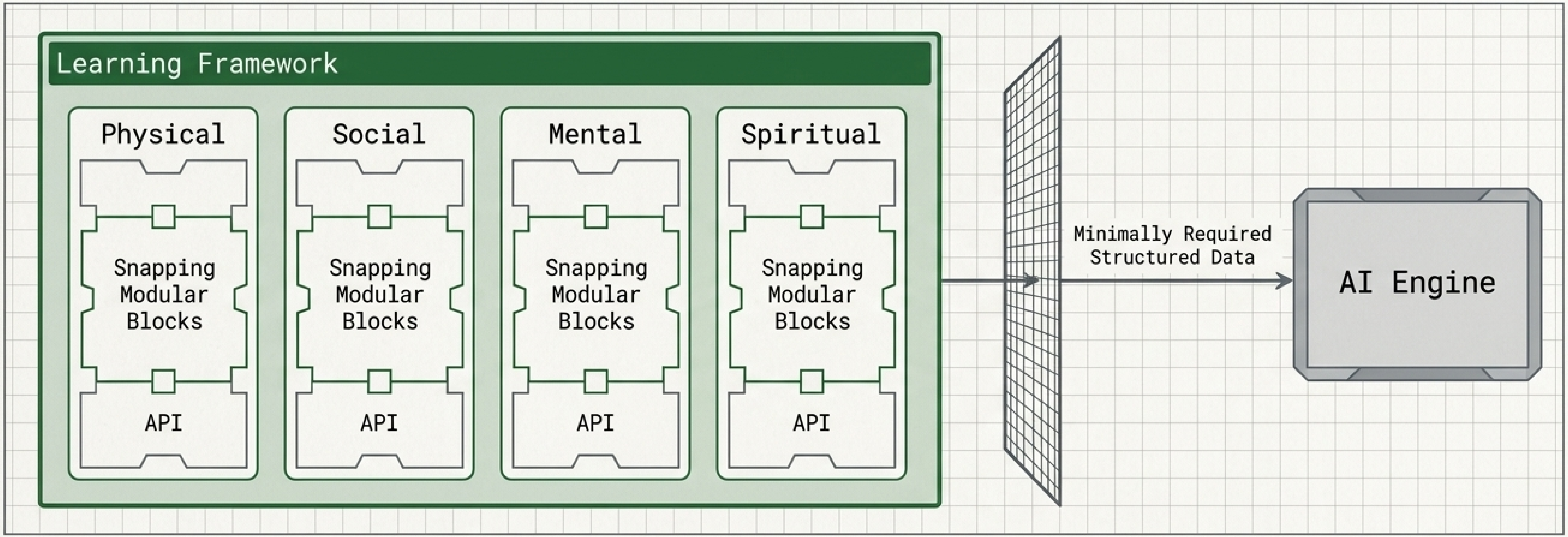
System Rule

State changes are tightly controlled. Invalid transitions are outright rejected.

Integration Requirement

Task completion must be durable within the Project DB before an integration event is sent to downstream capabilities like Recognition.

Learning Framework Isolates Professional Development



Core Mandate

Track learning activities and milestone completions across four professional dimensions.

System Rule

The framework records progress, but is strictly configured for professional development, making no medical or psychological claims.

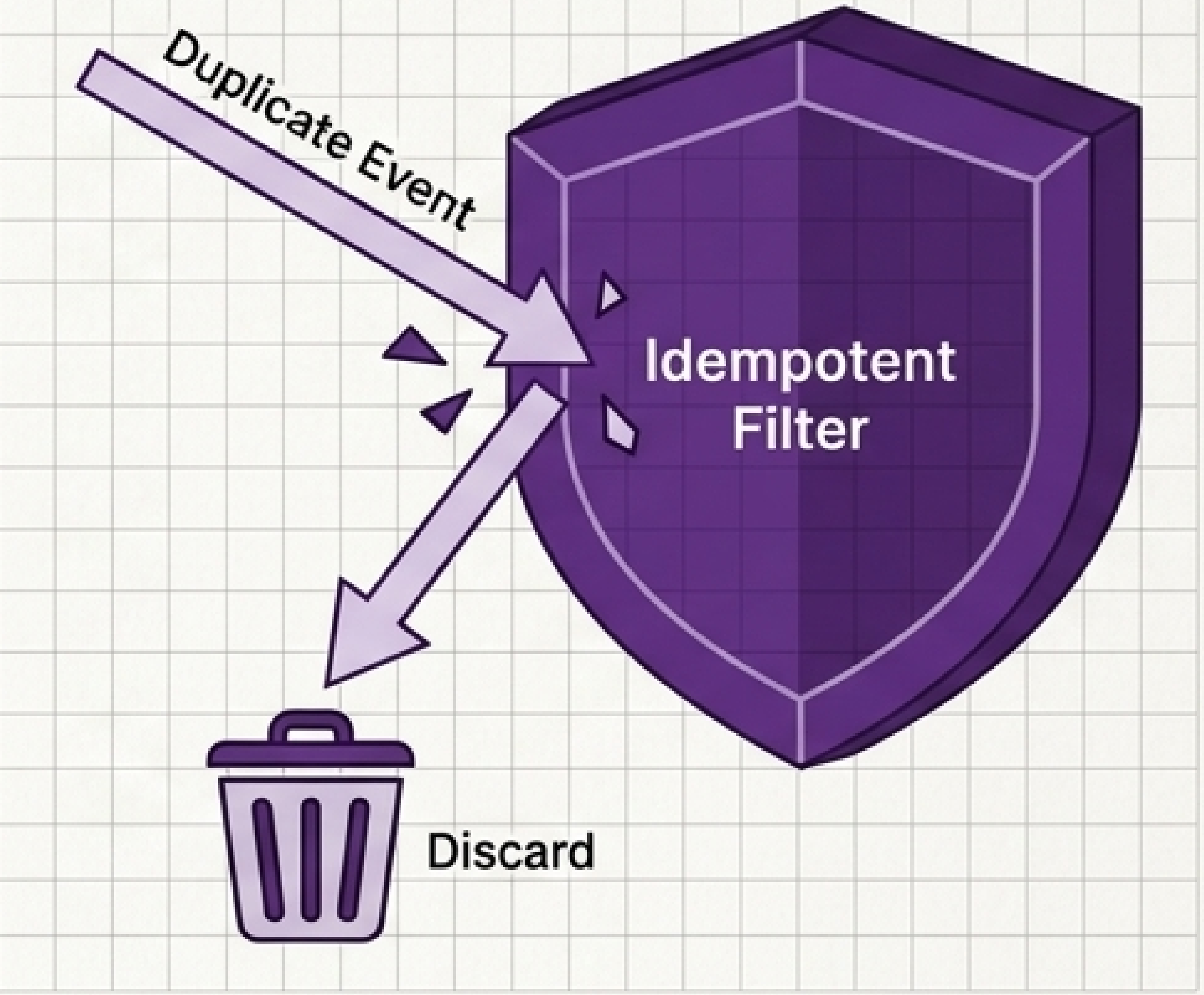
AI Boundary Constraint

AI acts only in an advisory recommendation capacity. It cannot corrupt learning progress or write directly to the Learning DB.

Recognition Ledger Guarantees Transactional Integrity

Recognition Ledger

	Timestamp (UTC)	Event Source	Points
1	2024-10-27T10:15:00Z	Task Completed (ID: TSK-101)	+50
2	2024-10-27T10:30:00Z	Milestone Reached (ID: MST-204)	+100
3	2024-10-27T11:05:00Z	Peer Recognition (ID: REC-309)	+25
4	2024-10-27T11:20:00Z	Workflow Approval (ID: WFA-505)	+75
5	2024-10-27T11:45:00Z	Goal Achievement (ID: GAL-801)	+150



Core Mandate

Maintain an auditable point and reward transaction ledger.

System Rule

Prevent duplicate point-changing operations. (FR-RECO-004)

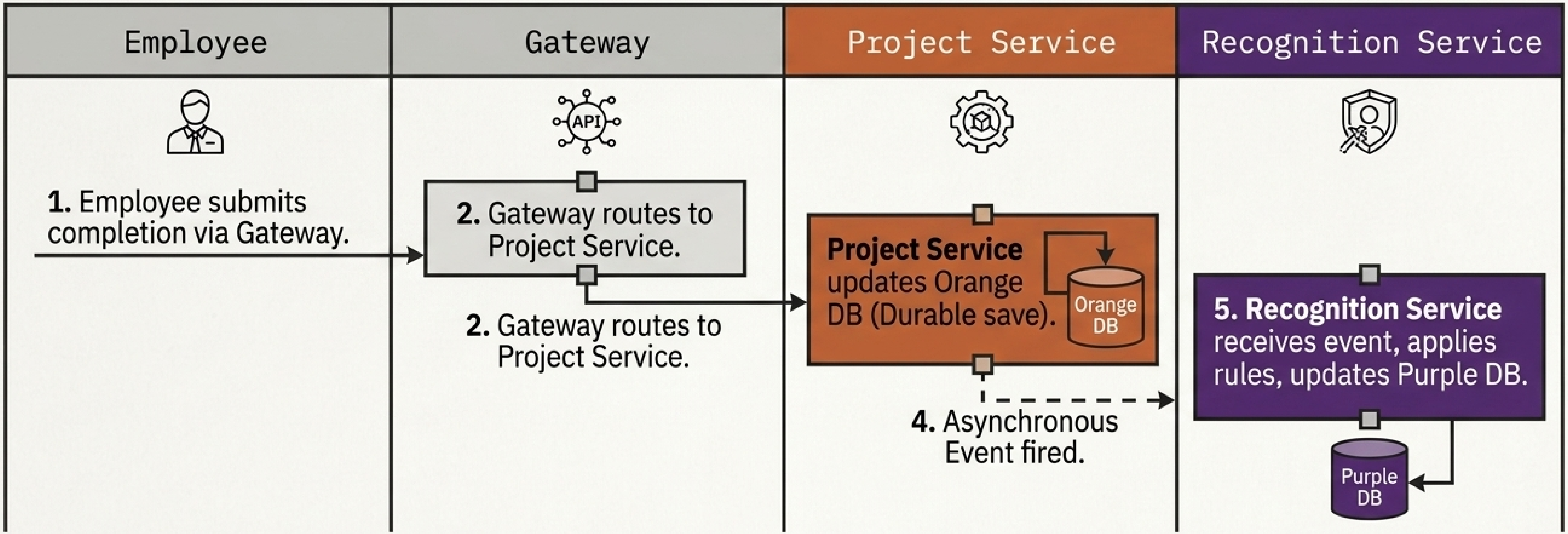
Design Pattern

Repeated completion deliveries from upstream services must be processed idempotently to ensure no invalid or duplicate point inflation occurs after network failures.

Decoupling Modules via Structured Communication

	Synchronous (REST)	Asynchronous (Events)
Use Case	**Immediate UI feedback required** (e.g., Client to Services).	**Independent downstream reactions** (e.g., Project triggering Recognition). Eventual consistency is acceptable.
Failure Handling	**Explicit timeouts**, controlled fallback errors, consistent validation rejections.	**Preserves valid originating task**; utilizes retry mechanisms and idempotent consumers. Broker must be free/open-source.

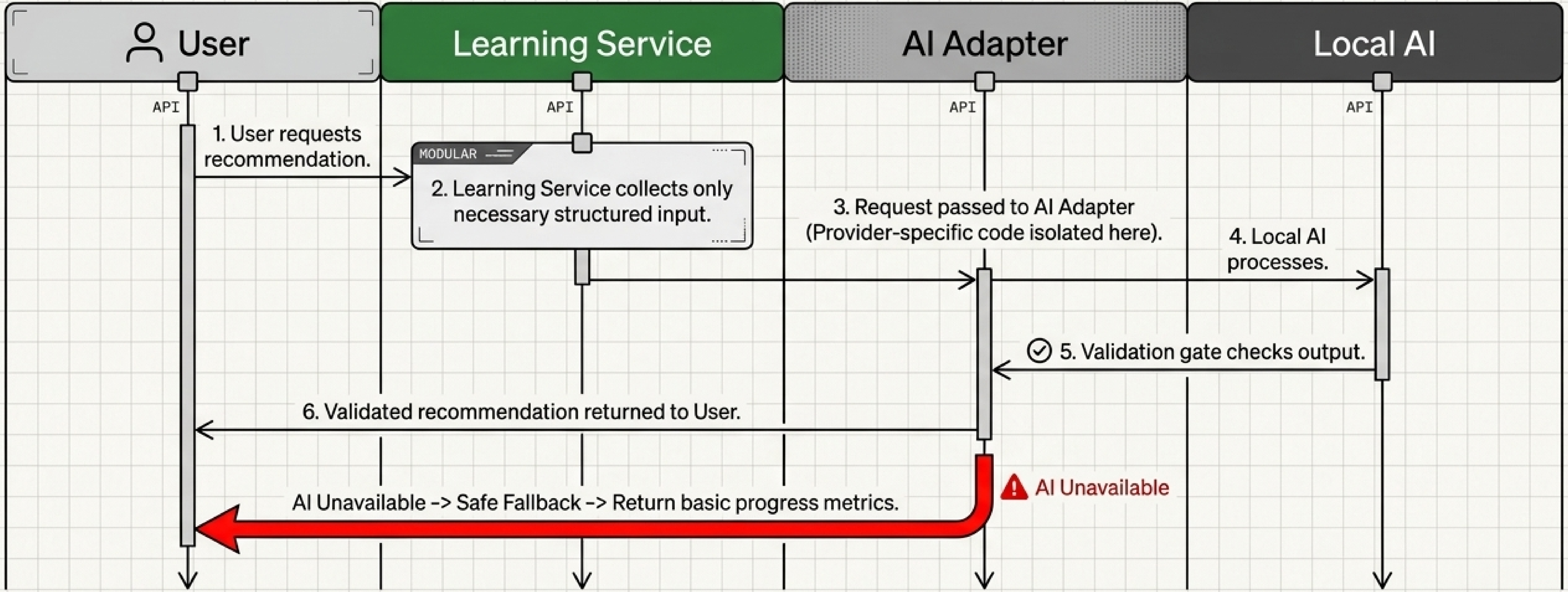
Tracing the Task-to-Recognition Event Workflow



Downstream failures (Recognition unavailable) never corrupt the source transaction (Project state).

The system favors eventual consistency over distributed locking to maintain high availability.

Confining AI to a Safe, Non-Destructive Loop



Core business state is **never** dependent on AI **availability.**
The AI Adapter ensures no provider-specific code leaks into the Learning domain.

Defining the System's AI Integration Boundaries

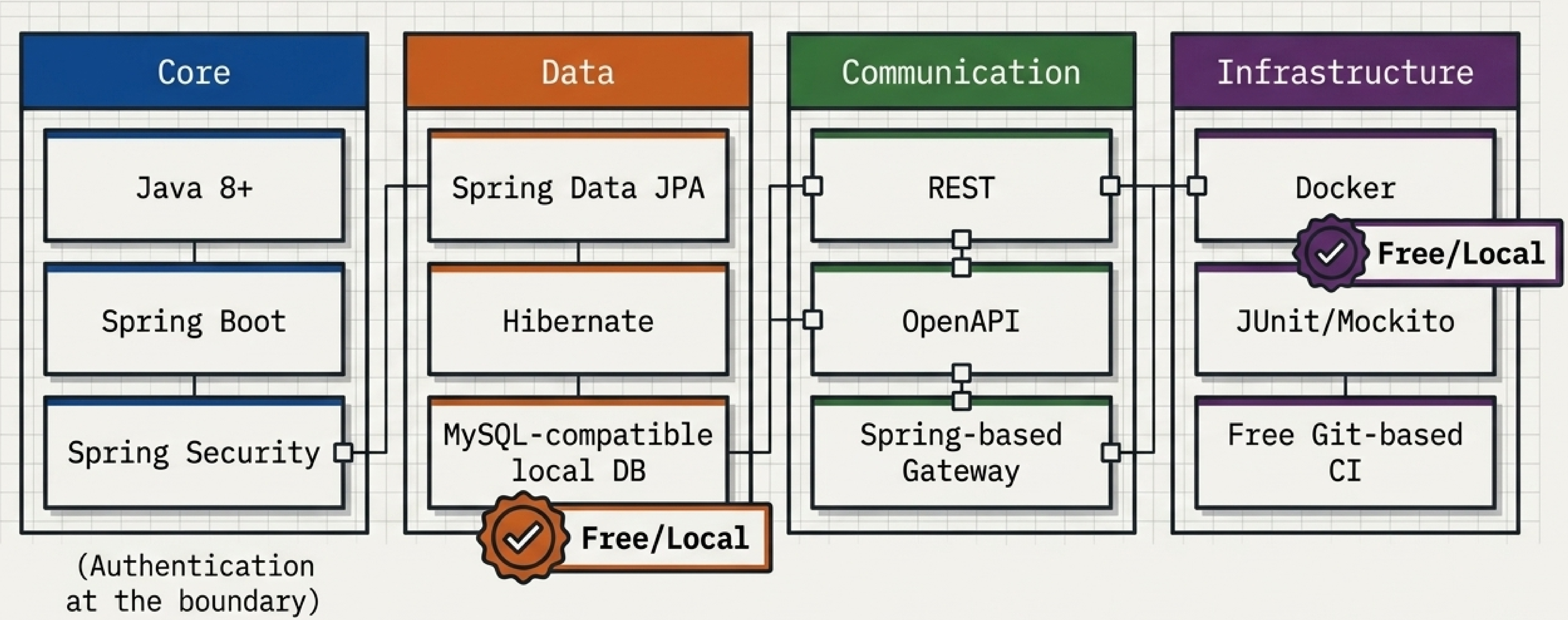
✓ Permitted AI Capabilities

- ✓ Advisory recommendations
- ✓ Structured data processing
- ✓ Local execution via free models
- ✓ Isolated provider-specific adapters

✕ Restricted AI Actions

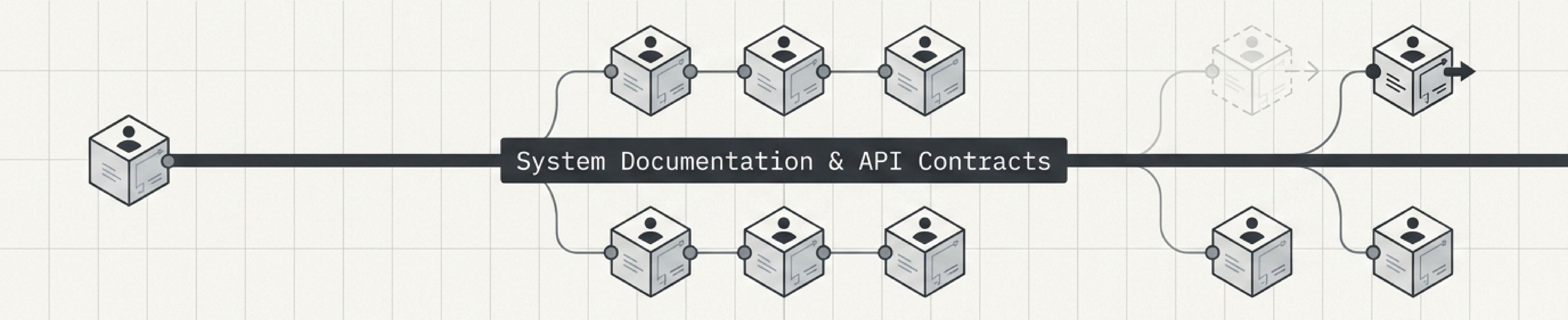
- ✕ Direct database modification
- ✕ Independent sensitive business actions
- ✕ Acting as a critical path dependency
- ✕ Mandatory paid API usage
- ✕ Storing shared core data

The Zero-Cost Java Technology Baseline



/* Note: No advanced infrastructure or paid cloud tools until organically justified. */

Guaranteeing Developer Continuity at Any Scale



1 to N Scaling

Because interfaces and data ownership are strictly drawn, one developer can work sequentially, or multiple developers can work simultaneously against stable API contracts.

Knowledge Retention

Documentation-first architecture ensures a developer can assume ownership of a module solely via repository READMEs and OpenAPI specs, eliminating undocumented tribal knowledge.